

Atmosphere TV

# DOOH Venue Incrementality & Media-Mix Measurement

*A causal measurement framework, calibrated media-mix model, budget optimizer, and venue revenue + retention models — built to answer the questions Atmosphere's advertising business and its own network economics run on.*

# How I'd approach this role

*Before the demo — the operating thesis behind it: what this role should do for Atmosphere, and how I'd prioritize the first few months as its first data scientist.*

## Advertising-side measurement

Prove Atmosphere's own incremental foot traffic with methods advertisers can trust — the sell-side differentiator this business runs on. (Answers Q1–Q2.)

## Venue-network health

Know which venues are generating less ad revenue than their audience is worth, and which are at risk of leaving — protect and grow the inventory Atmosphere actually sells. (Answers Q3–Q4.)

## Not a “quick win vs. foundation” tradeoff — do both in one move

*A stratified RCT geo-holdout is the fastest trustworthy result available — and it doubles as the calibration anchor every other method here relies on: synthetic control's placebo check, the MMM's scale, and the venue models' causal-value feature. Ship it first and the “quick win” and the “foundation” are the same deliverable. The venue-network data pipeline (economics + retention features) doesn't have to wait on it — it builds in parallel.*

## What “done” means

Sales and account teams making different decisions with it: a trusted lift number in a client pitch, a budget conversation anchored to a real response curve, an outreach list ranked by value × risk instead of instinct — not just a model with a good accuracy number.

*What follows is one concrete example of applying this approach — a synthetic demo.*

# Honest scope

*Before the demo that follows: the scope caveats specific to this project, stated up front rather than saved for the closing slide.*

## Synthetic data

Every figure in this project is synthetic. The focus here is the methodology for solving the problem, not real company data or figures.

## Media-effectiveness & rate-card figures

Dwell time, screen count, ad rate card, and similar parameters shown are illustrative demo values, not researched real industry benchmarks. In production: Nielsen OOH, DSP data (e.g. Vistar), or Atmosphere's own play logs and rate card.

## Multi-touch attribution — deliberately not built

Atmosphere's ambient-screen model has no individual-level, cross-venue touchpoint log by default. Building MTA would require purchased mobile location/device-matching data — a real but non-default assumption, so it's scoped out rather than forced.

## Online-purchase attribution — also out of scope

Every method here measures incremental foot traffic, not downstream online purchases. Tying exposure to e-commerce conversions needs a device-matched exposure-to-transaction panel — a further non-default assumption layered on top of MTA's.

## Cost & revenue assumptions

The \$/frequency-unit figures behind the budget allocator, and the ad-rate-card behind the venue-revenue model, are illustrative, editable placeholders — in production these come from Atmosphere's own rate card by venue type and daypart.

## Retention model — its own caveats

Churn-hazard, competitive-geo, and engagement-signal parameters are illustrative, not researched attrition benchmarks. geo\_cluster is deliberately excluded (small-sample tradeoff); realized\_ad\_revenue's importance is a venue\_type confound, not a causal driver; no prospect-side retention model.

# Four questions this project answers

Q1-Q2 answer the advertising side (what Atmosphere sells); Q3-Q4 answer the venue-network side (what Atmosphere runs).

1

## Did the campaign actually work?

Isolate the TRUE incremental foot traffic caused by ad exposure — net of seasonality, trend, and each venue's own baseline pattern.

The sell-side differentiator: measurement Atmosphere's go-to-market team can take to market and clients can trust.

2

## How should budget be spent?

Given a budget an advertiser has already committed to Atmosphere, how should it split across restaurants, gyms, bars, and waiting rooms — accounting for each venue type's own diminishing-returns curve?

An allocation Atmosphere optimizes on its own inventory and hands advertisers as part of the media plan — not a cross-channel budget call made elsewhere.

3

## Where's the revenue upside?

Which existing venues are under-monetized relative to their own traffic and quality — and which prospective venues are worth prioritizing for expansion?

Atmosphere's own buy-side value question.

4

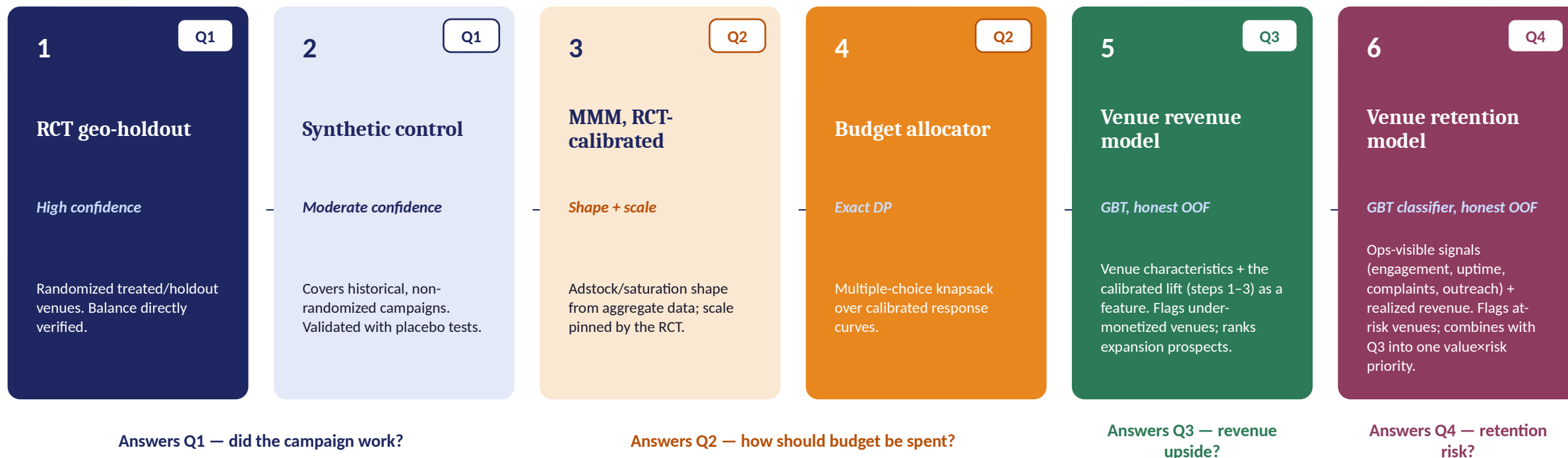
## Which venues are at risk?

Which existing venues are likely to leave the network, and which ops-visible signals predict it — combined with Q3's revenue into one value-x-risk priority.

Atmosphere's own buy-side risk question.

# The roadmap: four problems, one connected pipeline

Before the deep dive — this is the shape of what follows.



One connected system, not four separate projects: the calibrated per-exposure lift from steps 1-4 becomes a feature in the venue-revenue model (step 5), which combines with the churn model's risk score (step 6) into one value- $\times$ -risk priority list.

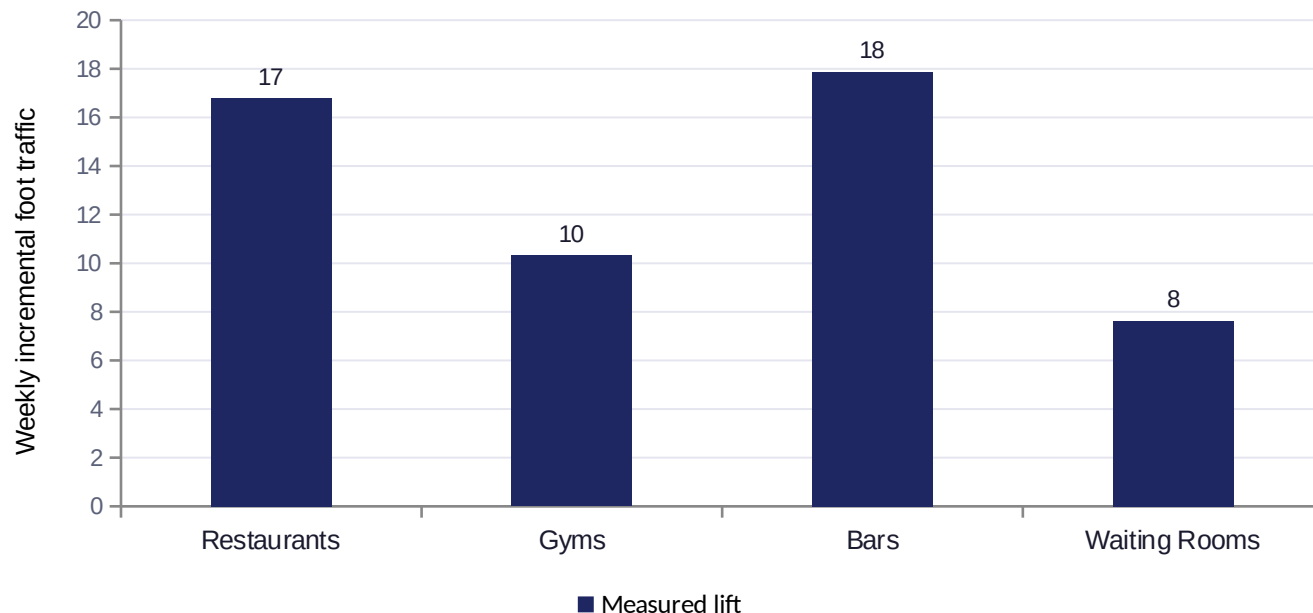
# Two data sources: a designed holdout test, and campaign history

4 venue types × 100 venues each, 104 weeks (52 pre-period + 52 campaign)

| Weeks                               | RCT pool — 136 venues                                                                         | Observational pool — 264 venues                                                            |
|-------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 0-51<br>(52 wks)<br>Pre-period      | No ads for either arm — establishes each venue's own baseline.                                | No ads yet for anyone — identical baseline period.                                         |
| 52-59<br>(8 wks)                    | Still silent — the designed test hasn't launched yet. Unused by the current effect estimate.  | 24 of 264 go live here — each on its own timing, no coordination.                          |
| 60-69<br>(10 wks)<br>RCT window     | Treated: fixed 6 plays/wk. Holdout: still 0. The only window the RCT estimate uses.           | 37 more go live here — overlap with the RCT window is coincidence, not design.             |
| 70-103<br>(34 wks)<br>Post-campaign | Ads stop — a real tail holds for ~10 weeks, then fades to zero (see persistence check, next). | 69 more (the largest group) go live only now — most real campaigns land late, spread thin. |

RCT pool: assignment is verified, not assumed — anchors high confidence. Observational pool: 134 of 264 venues never run a campaign at all, and the rest 130 go live at their own, uncoordinated point in time — whether a venue is using its own free self-promotion slot or Atmosphere sold the remaining inventory there to an outside advertiser, the panel only records that the screen started playing ads, not who bought the airtime — exactly why synthetic control (next) reconstructs the counterfactual instead of comparing before/after directly.

# The proof point: ads move real foot traffic



## Method: ANCOVA-style delta estimator

- ① Per-venue delta = campaign-window avg – pre-period avg — nets out that venue's own baseline.
- ② Effect = mean(treated deltas) – mean(holdout deltas) — a DiD-style comparison; any trend common to both arms cancels out too.
- ③ Welch's t-test on the two delta samples (unequal variance, unequal n) → SE, 95% CI, p-value.

Why delta, not raw post-period levels: individual venues differ a lot in baseline traffic even within one arm — differencing each against its own pre-period average removes that noise, so 15–19 venues/arm reach significance a raw-level comparison couldn't. Why Welch's, not Student's or z: arms are small and unequal in size, so we don't assume equal variance, and we use the t- (not z-) distribution — for both the p-value and the CI — because that variance is itself estimated from so few venues.

| Venue type    | n (T/H) | 95% CI       | p-value |
|---------------|---------|--------------|---------|
| Restaurants   | 18/17   | [10.9, 22.7] | < 0.001 |
| Gyms          | 15/17   | [5.0, 15.7]  | < 0.001 |
| Bars          | 17/19   | [13.1, 22.6] | < 0.001 |
| Waiting Rooms | 17/16   | [5.0, 10.3]  | < 0.001 |

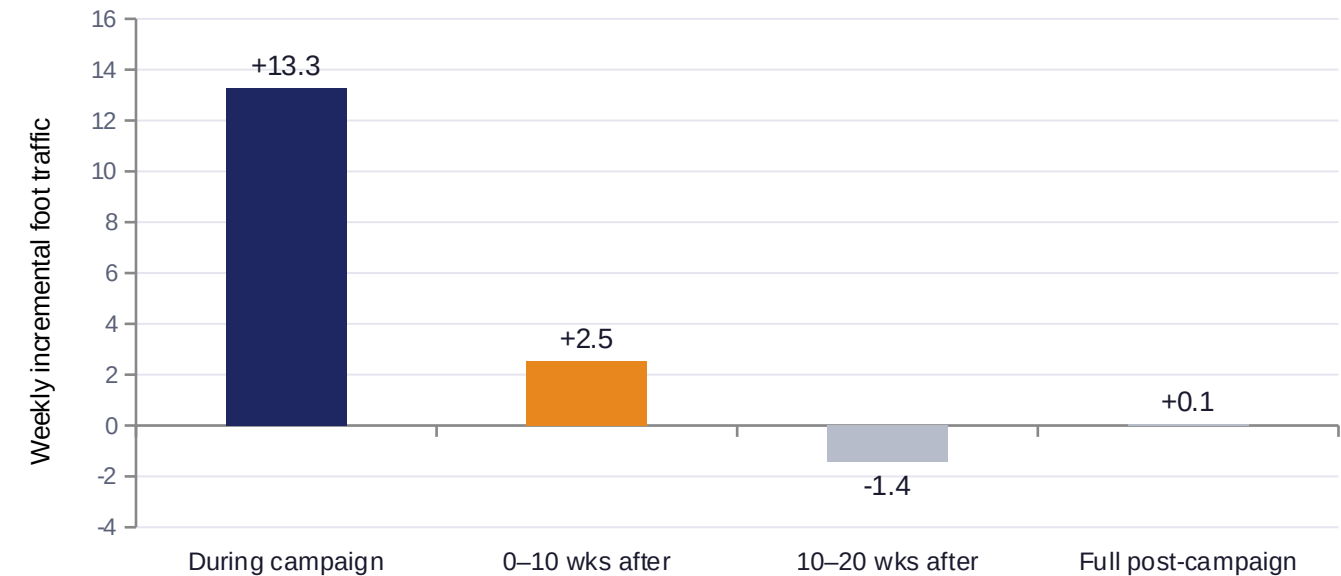
## Why 2/16 isn't a red flag

16 t-tests = 4 pre-treatment covariates (baseline traffic level, dwell time, screen count, audience quality) × 4 venue types — a separate check from the 136 RCT venues themselves. Only 2 came back significant at  $p \leq 0.05$ .

*If randomization is clean, the count of false positives across 16 independent t-tests follows Binomial(16, 0.05): mean 0.8,  $P(\geq 2) \approx 19\%$ . Two sits well inside that distribution's main mass — nowhere near a red-flag tail value like 8+. Randomization worked as designed, not assumed.*

# How long does the lift last after the campaign ends?

Same treated-vs-holdout contrast as the previous slide, applied to the weeks after the campaign ends — no new design, no new assumption.



## Method: same estimator, later windows

Same 3-step ANCOVA-style delta as the RCT slide — per-venue delta (window avg – pre-period avg) → treated-mean minus holdout-mean → Welch's t-test for SE, CI, p-value — but pooled across all 4 venue types here (not split like the RCT slide), and re-applied to each window instead of the single week 60-69 window.

Splitting was already thin at ~15-19 venues/arm for one window; spreading that across four venue types and four windows, on top of decay effects that are smaller than the in-campaign lift, would push the noise past what a week-level estimate can resolve.

n = 67 treated / 69 holdout (same venues, every window)

| Window           | 95% CI       | p-value |
|------------------|--------------|---------|
| During campaign  | [10.8, 15.8] | < 0.001 |
| 0-10 wks after   | [0.2, 4.8]   | 0.031   |
| 10-20 wks after  | [-3.8, 0.9]  | 0.236   |
| Full post (34wk) | [-1.4, 1.5]  | 0.945   |

## A small tail, then nothing

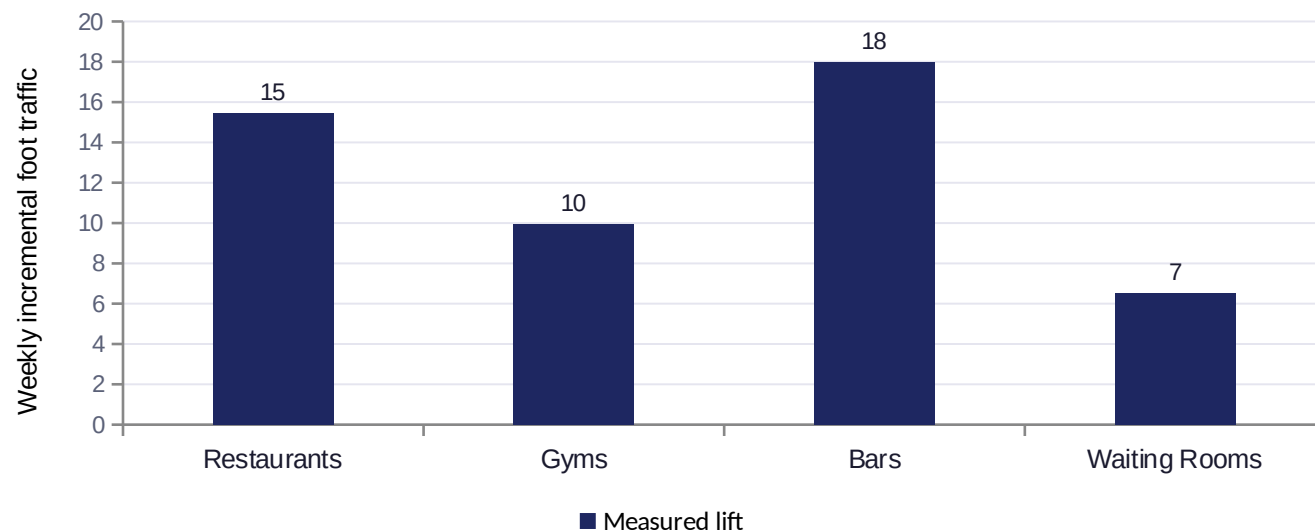
The first 10 weeks after the campaign ends still show a real, significant tail — about 19% of the in-campaign lift (p=0.031). By 10-20 weeks out it's gone (p=0.236), and averaged across the full 34-week post-campaign panel the effect is a clean +0.05 (p=0.945).

Practical read: don't plan on meaningful carryover past the campaign's own window — each flight buys its own 10 weeks of impact, not lingering awareness. That argues for cadence, not one-and-done bursts.



# Measuring the campaigns we couldn't randomize

Most advertisers' campaigns already ran, on venues they picked — not us. Synthetic control reconstructs the counterfactual anyway, with its own built-in checks.



| Venue type    | n(used/flag) | Placebo p | RMSPE |
|---------------|--------------|-----------|-------|
| Restaurants   | 42/4         | 0.158     | 26.4  |
| Gyms          | 31/0         | 0.054     | 14.4  |
| Bars          | 29/4         | 0.097     | 17.8  |
| Waiting Rooms | 20/0         | 0.106     | 8.3   |

Pre-period RMSPE (root mean squared prediction error) =  $\sqrt{\text{mean}((\text{actual} - \text{synthetic})^2)}$  over each venue's own pre-period weeks — how well the twin matches reality before treatment starts. Flagged (excluded from the bar) if RMSPE exceeds 2.5× its type's median — restaurants' high median (26.4) is why 4 of its 46 fits got flagged.

## Method: build a synthetic twin, per venue

- ① Donor pool = same-type venues that never ran a campaign (e.g. 31 never-activated bars).
- ② Fit non-negative weights, one per donor, summing to 1, via SLSQP (Sequential Least Squares Programming, `scipy.optimize`) minimizing the MSE between the weighted donor blend and this venue's own pre-period traffic (Abadie et al.) — a real blend, typically 5–15 donors get meaningful weight, not one single match.
- ③ Apply that same weight vector to the donors' post-period traffic → synthetic counterfactual; effect =  $\text{mean}(\text{actual} - \text{synthetic})$  over the venue's own 16-week post-campaign window.

Unlike the RCT, each venue picks its own pre/post split at its own activation week — no shared campaign window. Each bar is the mean effect across only that type's good pre-period-fit venues (see right) — poor fits are excluded, not averaged in.

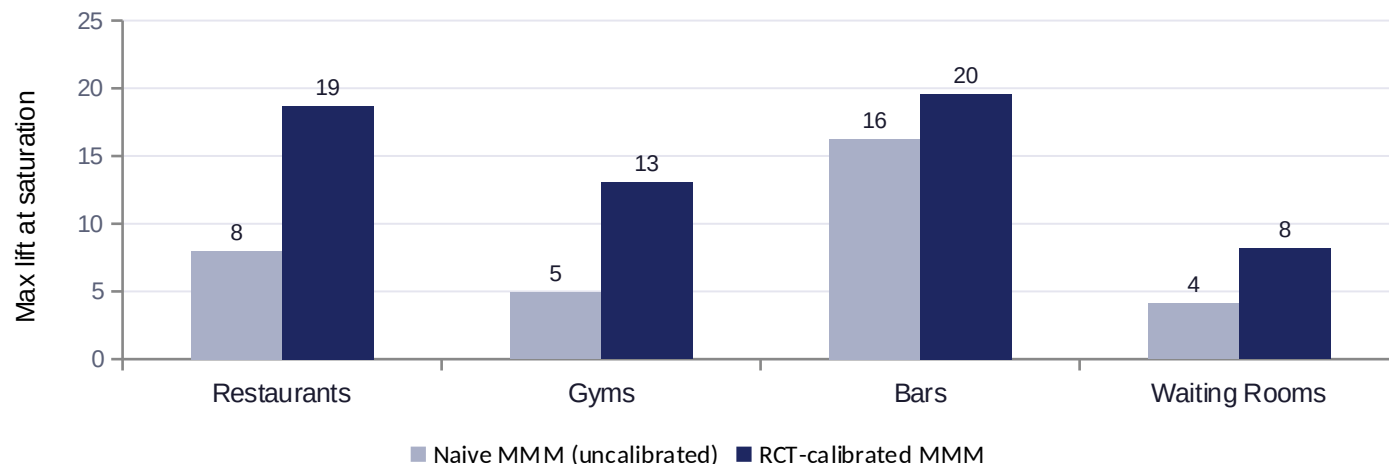
*In-space placebo test* — substitutes for a p-value since synthetic control has no closed-form SE: rerun the identical fit on every never-activated donor, pretending each was treated. Placebo p = share of donor "fake effects" at least as extreme as the real one — restaurant's p=0.158 is the weakest signal (its worst-fitting type too); gym's p=0.054 the strongest.

## Consistent with the RCT — not independently significant

No type clears  $p < 0.05$  alone (gyms closest, 0.054) — but every point estimate lands within ~1.3 of the RCT's own effect for that same type, on a completely different set of venues: real corroboration, just not from this test alone.

# MMM: an uncalibrated model would have underpriced results

Adstock + saturation shape comes from the aggregate weekly series; scale is pinned by the RCT's high-confidence estimate.



## Calibration adjustment

Restaurants **+134%**

Gyms **+165%**

Bars **+20%**

Waiting Rooms **+100%**

## Method: fit the shape, then pin the scale

① Fit shape on the aggregate weekly series (all venues of this type, RCT + observational pooled together): adstock carries exposure forward week to week ( $\text{adstock}_t = \text{freq}_t + \text{decay} \times \text{adstock}_{t-1}$ ), then a Hill curve turns that into diminishing returns ( $\text{sat}(x) = x^{\text{shape}} / (x^{\text{shape}} + \text{half}^{\text{shape}})$ ). decay, half, and shape are grid-searched —  $7 \times 6 \times 4 = 168$  combinations — picking whichever minimizes squared error against actual weekly traffic (with trend + seasonality controls, plain OLS). This gives  $\beta_{\text{naive}}$ , the raw uncalibrated max-lift.

② Pin scale only, via the RCT: simulate the RCT's own exposure (6 plays/wk for 10 weeks) through this same fitted decay/saturation curve to get its implied saturation level, then rescale  $\beta$  so the model's predicted lift there matches the RCT's point estimate exactly. decay/half/shape stay exactly as fit in step ① — only  $\beta$  moves.

## Worked example: restaurant's $\beta$ , step by step

RCT design: 6 plays/wk  $\times$  10 wks

Adstock (decay .7): 6.0  $\rightarrow$  10.2  $\rightarrow$  13.1  $\rightarrow$  ...  $\rightarrow$  19.4

Saturation (half 6, shape 3): .50  $\rightarrow$  .83  $\rightarrow$  .91  $\rightarrow$  ...  $\rightarrow$  .97, avg  $\approx$  0.90

Naive predicts:  $7.97 \times 0.90 \approx 7.2$

RCT measured: 16.79 (real, from Slide 7)

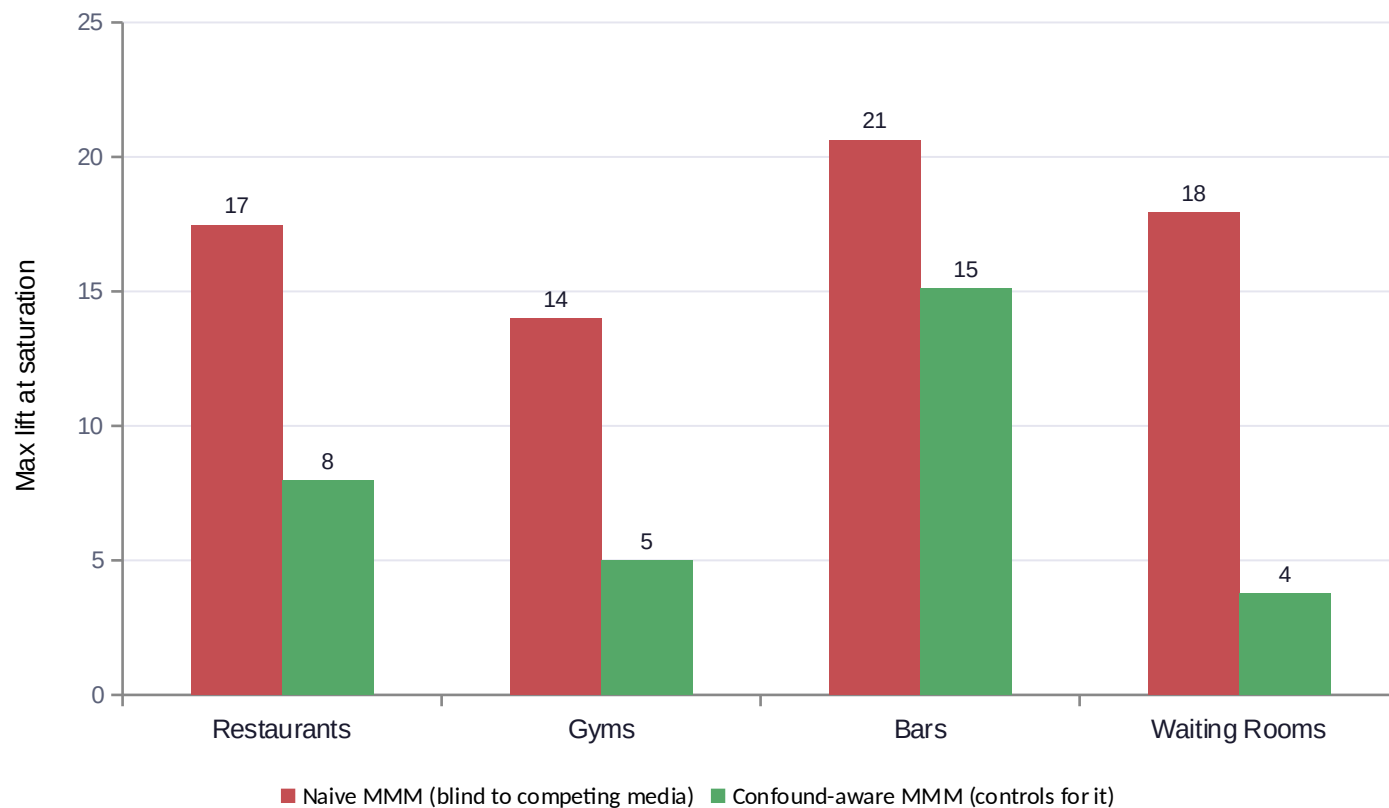
Calibrated  $\beta$ :  $16.79 \div 0.90 \approx 18.7$

Only  $\beta$  is rescaled — decay/half/shape stay exactly as fit in step ①. calibration\_adjustment\_pct is just  $(\beta_{\text{calibrated}} - \beta_{\text{naive}}) \div \beta_{\text{naive}}$ :  $(18.69 - 7.97) \div 7.97 \approx +134\%$ , the number shown for restaurants above.

*Simplified for the demo: a real deployment would add more controls (competitor spend, price/promo, macro trend), validate the shape out-of-time rather than on in-sample SSE alone, and often blend the experiment in as a Bayesian prior rather than a hard rescale.*

# Robustness check: does competing media break this MMM?

A stress-test scenario: a competing advertiser ramps up on the same venues at the same time, correlated with Atmosphere's own exposure — the kind of overlap a real MMM has to prove it's robust to.



## RCT estimate shift under the identical shock

0.00

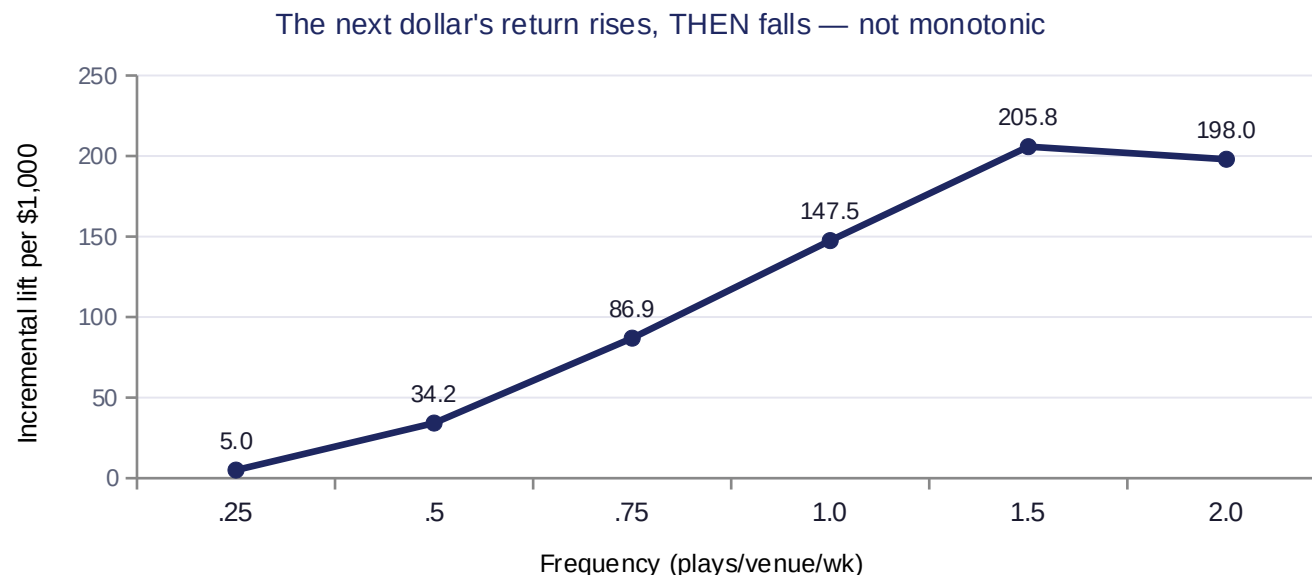
every venue type — the shock hits both arms equally, so a randomized comparison is mechanically immune to it

Naive MMM, by contrast, swings from useful to actively misleading: for 3 of 4 venue types the confound-blind fit was off by 2–4× — even at a moderate 0.35 correlation between competing media and Atmosphere's own exposure ramp, nowhere near a worst case. Controlling for the confound (green) recovers this model's own baseline from the previous slide — the RCT-scale calibration from that same slide is what closes the rest of the gap.

Why this is scoped this way: deciding how an advertiser splits budget across Atmosphere + TV + social is the advertiser's/agency's call — Atmosphere has no visibility into competitors' channel data anyway. What Atmosphere's DS role should own is netting competing media OUT of its own causal read (the JD's own phrasing), which the RCT does by design and a single-channel MMM only does with an explicit control.

# Why a greedy allocator can lose to equal-split

A Hill/S-curve (shape  $> 1$ , as fit here) is convex before its inflection point — the next dollar's marginal return isn't monotonically falling, so "chase today's best marginal \$" can get stuck.



This real curve (restaurant, from this deck's own calibrated MMM) is convex up to ~freq 1.0–1.5, concave after. At very low frequency every venue type's next dollar buys almost nothing — still in the flat, convex trough — so a rule comparing "which type's next dollar looks best right now" can't see that committing a lump sum to ONE type would pay off later. It only sees today's small, similar-looking numbers.

## At freq $\approx .25$ , every type looks similarly unpromising

| Venue type   | Lift per \$1,000 |
|--------------|------------------|
| Bar          | 1.4              |
| Restaurant   | 5.0              |
| Gym          | 6.1              |
| Waiting room | 8.4              |

...but each type's own PEAK (always somewhere around freq 1–2) is 15–30× higher:

|          |                   |
|----------|-------------------|
| Bar ~124 | Restaurant ~206   |
| Gym ~132 | Waiting room ~247 |

A greedy rule that only ever compares "today's next dollar" has no way to see that peak from here — it can only react to which of these near-tied numbers is currently largest, one tiny step at a time. This is the mechanism that let an earlier greedy version of this allocator underperform naive equal-split at larger budgets — the retired code itself isn't preserved, but this is the exact response-curve shape it was walking.

# Budget allocator: how the exact DP actually works

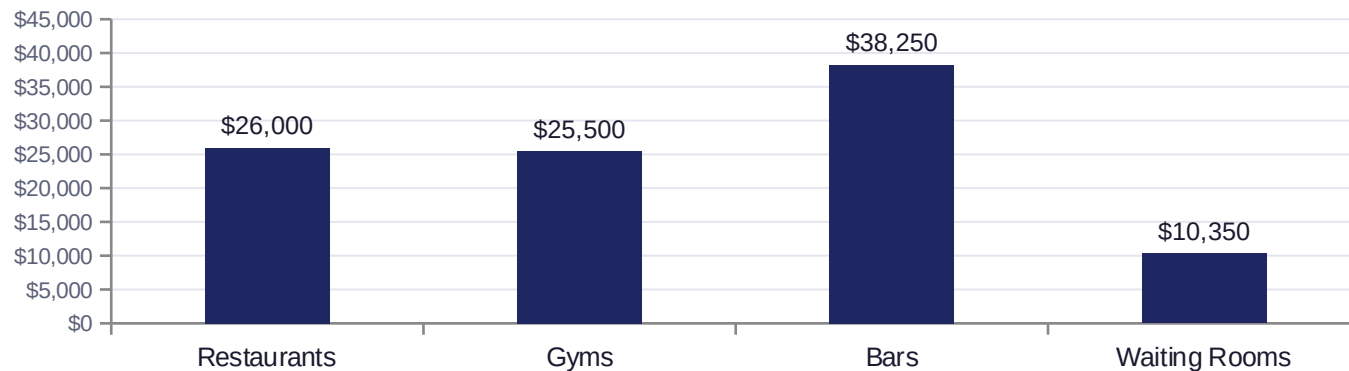
No concavity requirement — every discretized spend level is actually evaluated, so by construction it can never do worse than equal-split.

## Method: exact DP in four steps (multiple-choice knapsack)

- ① Discretize: for each venue\_type, sweep frequency  $0 \rightarrow 14/\text{venue}$  in 0.25 steps (57 points). At each point, read network lift = per-venue lift (from the calibrated MMM response curve)  $\times$  venue count, and \$ spend = freq  $\times$  cost-per-frequency-unit  $\times$  venue count.
- ② Snap every candidate's spend onto a shared \$250 grid, so all venue types' options line up on one budget axis — keeping only the highest-lift option at each grid point per type. This is that type's "menu."
- ③ Process venue types one at a time: for every \$ grid point  $b$ , try each of that type's menu options with spend  $\leq b$ , add it to whatever the types-already-processed achieved at  $(b - \text{spend})$ , and keep whichever combination gives the highest total lift — every option is actually tried, not estimated from a local slope.
- ④ Backtrack from the full-budget column to read off which frequency was actually chosen for each venue type.

Grid granularity (\$250 budget steps, 0.25 frequency steps) trades a little precision for tractability — the same principle real deployments use, just at production-scale resolution.

\$100,000 weekly budget — DP-optimal allocation



## Gain vs. naive equal-split

\$20,000 budget **+48.6%**

vs. naive equal-split

\$50,000 budget **+11.1%**

vs. naive equal-split

\$100,000 budget **+2.4%**

vs. naive equal-split

The optimizer's edge is largest when budget is scarce — exactly when allocation decisions matter most. (Previous slide: why a greedy walk can lose this edge entirely.)

# The other side of the business

*One system answers the advertising side. The same system answers the venue-network side too.*

## Advertising incrementality

- Prove and price advertising incrementality
- Feeds go-to-market: sell-side differentiator, client trust
- RCT + synthetic control + calibrated MMM + budget DP

*Feeds the venue-economics model as a validated causal-value input (calibrated per-exposure lift), not a separate silo.*

## Venue network economics

- Predict realized ad-revenue (Q3) and 90-day churn risk (Q4) from characteristics and ops signals
- Flag under-monetized (Q3) and at-risk (Q4) existing venues for sales/ops follow-up
- Rank prospective venues for expansion priority (Q3)
- Combine Q3 revenue + Q4 risk into one value-x-risk priority quadrant
- GBT regressor + classifier, honest out-of-fold evaluation throughout

*“Smarter on both sides of the business” — the venue-side model shares its data foundation and causal-value input with the advertising side, rather than being a disconnected second project. Results on the next slide.*

# Venue revenue model: what feeds in, and why this model

*Can this venue's own characteristics + a validated advertising effect predict what it actually earns — and surface who's under-monetized or high-potential?*

## Model: HistGradientBoostingRegressor (sklearn)

Chosen because it handles venue\_type / geo\_cluster / traffic\_tier as native categorical splits. geo\_cluster alone has 20+ levels — one-hot encoding that would blow the feature space up and dilute each split's signal. categorical\_features="from\_dtype" avoids that entirely, no preprocessing needed.

## Target: realized\_ad\_revenue

*The venue's actual weekly \$ ad revenue — not a proxy, not a survey response.*

## Features — two sources

### This venue's own characteristics

venue\_type · geo\_cluster · traffic\_tier  
baseline\_level · dwell\_time\_min  
screen\_count · audience\_quality

### From the Q1/Q2 causal pipeline (RCT: Slide 7 • MMM: Slide 10)

calibrated\_lift\_feature — the RCT-calibrated per-exposure lift from the MMM. This is where the advertising-incrementality pipeline plugs into the venue-economics side — the connection promised on Slide 14, not a separate silo.

*Method, honestly: held-out metrics (next slide) come from a genuine 30% test split, stratified by venue\_type — never train-set fit. Under-monetization flags come from 5-fold out-of-fold predictions — every venue's flag uses a model that never saw its own revenue. Prospect venues (no revenue history) are scored by that same trained model, not a separate heuristic.*

# Predicting venue ad-revenue, honestly

Held-out evaluation, feature importance, and the venues these numbers actually flag (model & features: previous slide).

0.84

Held-out R<sup>2</sup>

22.1%

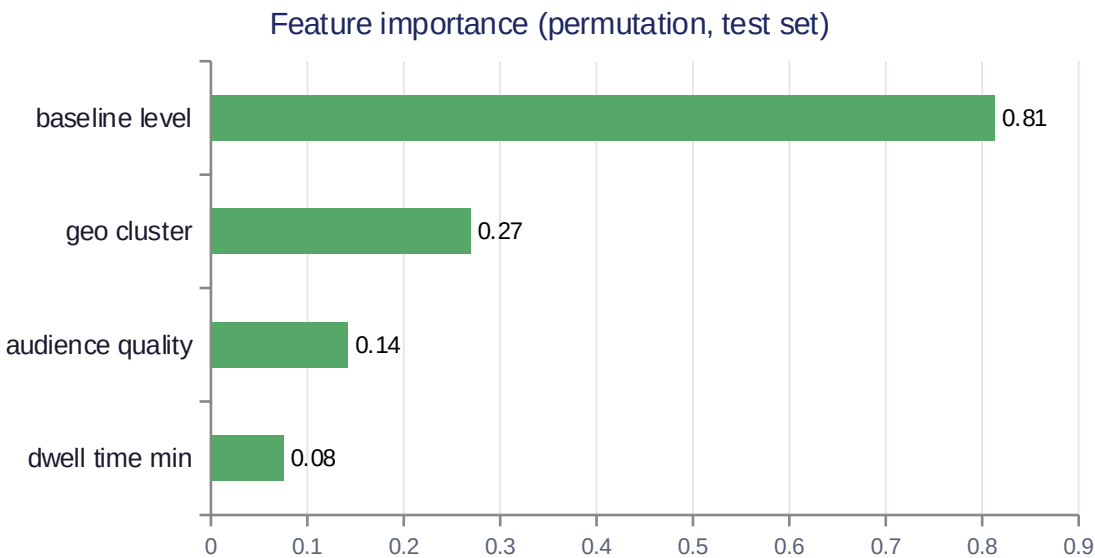
Held-out MAPE

\$45

Held-out MAE

20

Venues flagged



Permutation, not gain-based — gain-based splits are biased toward high-cardinality categoricals like `geo_cluster` (20+ levels).

Flags come from 5-fold out-of-fold predictions, never a model scoring the venue it was trained on. The calibrated-lift feature carries near-zero importance — honest, since it's constant within `venue_type` once `venue_type` itself is a feature.

## Top under-monetized venues (of 20 flagged)

$gap\_pct = (actual - OOF\ predicted) \div OOF\ predicted \times 100$

|                            |      |
|----------------------------|------|
| Venue #85 (Restaurants)    | -62% |
| Venue #309 (Waiting Rooms) | -61% |
| Venue #53 (Restaurants)    | -61% |
| Venue #247 (Bars)          | -60% |

## Top prospect venues (expansion priority)

|                                       |       |
|---------------------------------------|-------|
| P0018 — Bars (existing market)        | \$684 |
| P0055 — Restaurants (existing market) | \$524 |
| P0000 — Bars (new market)             | \$500 |
| P0001 — Bars (new market)             | \$496 |



# Predicting venue churn risk, honestly

Gradient-boosted classifier on ops-visible signals (engagement, uptime, complaints, competitor outreach) — the other half of "what makes a venue valuable and what puts it at risk."

0.65

Held-out AUC

0.27

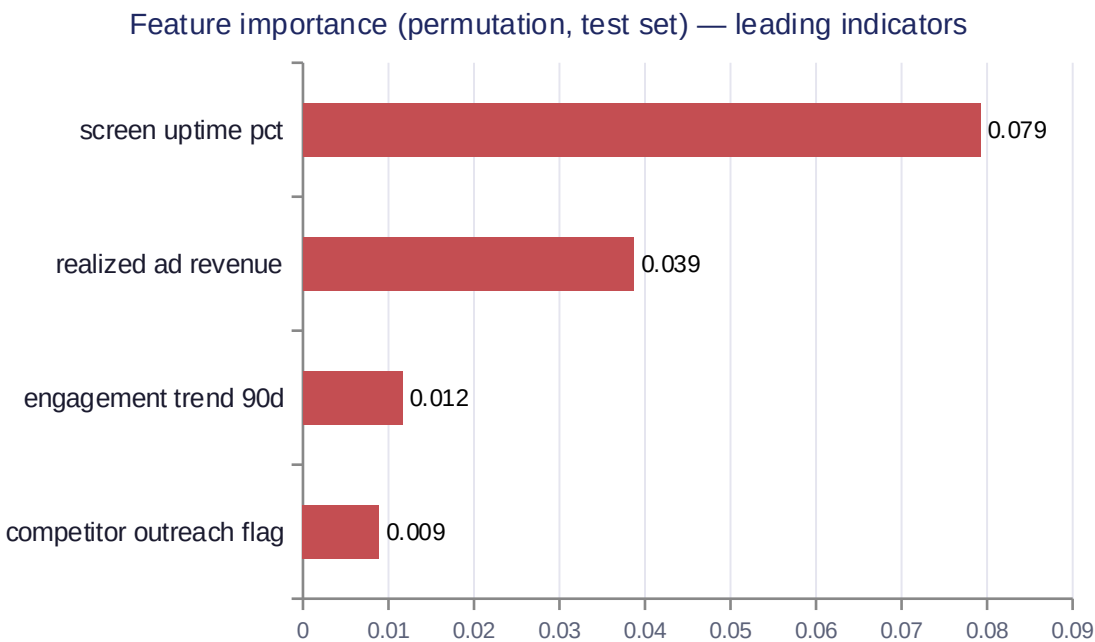
Held-out PR-AUC

0.13

Brier score

20

Venues flagged



## Top at-risk venues (of 20 flagged, by OOF risk)

|                                                 |     |
|-------------------------------------------------|-----|
| Venue #281 (Bars)                               | 60% |
| engagement declining → content/placement review |     |
| Venue #200 (Bars)                               | 55% |
| engagement declining → content/placement review |     |
| Venue #12 (Restaurants)                         | 52% |
| uptime issue → dispatch technical support       |     |

## Priority quadrant (Q3 revenue × Q4 risk)

|                                     |     |
|-------------------------------------|-----|
| Save now (high value, high risk)    | 136 |
| Monitor (low value, low risk)       | 136 |
| Protect (high value, low risk)      | 64  |
| Low priority (low value, high risk) | 64  |

Flags come from 5-fold OOF predictions; AUC/PR-AUC read modest next to the revenue model's  $R^2$  because churn is a rarer, noisier event to call than revenue — typical for a churn classifier, not a modeling miss. realized\_ad\_revenue's importance is a venue\_type confound, not a causal driver. Each "Save now" / flagged venue also carries a SHAP top driver (from the fold model that never saw it) mapped to a concrete action — technical dispatch, AM retention call, content review — not a generic "reach out."

# Takeaways

- 1 Match the method to the assignment mechanism — RCT where randomization is designed, synthetic control where it isn't, each with its own validation check.
- 2 An uncalibrated MMM understated the RCT-calibrated estimate by up to 165% here — experiment calibration isn't optional polish, it's the identification fix.
- 3 Optimization needs the right algorithm for the curve's shape — a greedy heuristic silently lost to a naive baseline on non-concave response curves; the exact DP formulation doesn't.
- 4 One connected system, not four separate projects: the venue-revenue model reuses the causal pipeline's per-exposure value as an input; the retention model then combines its own churn-risk score with that revenue prediction into one value- $\times$ -risk priority quadrant — both models are evaluated honestly out-of-fold, never scored on a venue they were trained on, before either is trusted.

*Thank you*